

D686

Operating Systems for Computer Scientists

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A web server built by a six-person startup handles each request in a separate thread of one process. One thread corrupts a shared in-memory cache, and other threads immediately read the bad data. The server runs on a 32-core machine. Why can the other threads see the corruption?

- A. Threads never share anything but CPU time.
- B. Threads of the same process share one address space, including the heap.
- C. The 32 cores share one cache.
- D. Each thread has its own copy of memory that syncs automatically.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A web browser runs each tab as a separate process instead of as a thread. What is the main benefit?

- A. Lower total memory use.
- B. Faster data sharing between tabs.
- C. A crash or fault in one tab cannot corrupt another tab's memory.
- D. Tabs can never crash.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Three jobs arrive at time 0 in the order P1 (burst 8), P2 (burst 4), P3 (burst 2). The scheduler is non-preemptive shortest-job-first. The machine uses a fast SSD. What is the average waiting time?

- A. About 6.67
- B. About 7.33
- C. About 2.67
- D. 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Under first-come, first-served scheduling, jobs P1 (burst 8), P2 (burst 4), and P3 (burst 2) arrive at time 0 in that order. What is P3's waiting time?

- A.** 14
- B.** 2
- C.** 0
- D.** 12

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An administrator sets a round-robin time quantum of 1 ms on a system where each context switch takes 0.5 ms. The machine hosts shells for 200 users. What is the main effect?

- A.** Round robin behaves like FCFS.
- B.** The effect depends on the 200 users' typing speed.
- C.** Throughput improves, because a smaller quantum is always better.
- D.** About a third of CPU time is lost to context switching.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Job A holds a lock on the Orders table and waits for Customers; Job B holds Customers and waits for Orders. Both run nightly at 2 a.m. on a server with 64 GB of RAM. The fix requires every job to take locks in the same fixed order. Which deadlock condition does this break?

- A.** Mutual exclusion.
- B.** Circular wait.
- C.** None; adding RAM is the fix.
- D.** Hold and wait.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A system requires every process to request all the resources it will need at once, before it starts. Which deadlock condition does this remove?

- A.** All four conditions at once.
- B.** Circular wait.
- C.** Hold and wait.
- D.** No preemption.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A system uses 4 KB (4,096-byte) pages and has 16 GB of RAM. What are the page number and offset for logical address 13,000?

- A. Page 3, offset 712.
- B. Page 712, offset 3.
- C. Page 3, offset 13,000.
- D. Page 4, offset 3,384.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

With 1,024-byte pages, page 5 maps to frame 9. What physical address matches offset 100 in page 5?

- A. 9,316
- B. 5,220
- C. 9,100
- D. 9,216

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

After an administrator raised concurrent batch jobs from 10 to 40, CPU use fell from 70% to 15% and the disk is busy almost entirely with paging. The server came from a reputable vendor last year. What is happening, and what is the fix?

- A. Thrashing; reduce the number of concurrent jobs or add memory.
- B. Contact the vendor about a hardware fault.
- C. Buy a faster CPU.
- D. The CPU is underused, so add more jobs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Two threads each run $\text{balance} \leftarrow \text{balance} + 100$ on a shared account 1,000 times with no locking. The expected total increase is 200,000, but the actual increase is lower. The app runs on a four-core laptop. What is the fix?

- A.** Never use threads for financial data.
- B.** Add a short sleep between updates.
- C.** Run it on a machine with more cores.
- D.** Protect the update with a mutex so only one thread runs it at a time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A print-spooler service runs with full administrator rights on every workstation, though it only needs to write to the print-queue folder. The service vendor has been in business for 20 years. What is the best control?

- A.** Keep administrator rights but set a stronger password.
- B.** Run it under a dedicated low-privilege account that can reach only the print-queue folder.
- C.** Keep administrator rights because the vendor is trusted.
- D.** Disable every service on every workstation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A web server ~~built by a six person startup~~ handles each request in a separate thread of one process. One thread corrupts a shared in-memory cache, and other threads immediately read the bad data. ~~The server runs on a 32-core machine.~~ Why can the other threads see the corruption?

Target word: why corruption spreads **Decides:** All threads are in one process.

A	Absolute language	Threads share memory by design.
B	Correct	The cache lives in memory every thread can reach.
C	Camouflage	Hardware caches are not the application's cache.
D	Inverted operation	That describes processes more closely, and there is no automatic sync.

2. Answer C Concept 1 · transfer

A web browser runs each tab as a separate process instead of as a thread. What is the main benefit?

Target word: benefit of processes — reversed **Decides:** Processes have isolated memory.

A	Confusion set	Separate processes usually use more memory.
B	Inverted operation	Threads share faster; processes need inter-process communication.
C	Correct	Process isolation contains failures.
D	Absolute language	They can; the crash just stays contained.

3. Answer C Concept 2

Three jobs arrive at time 0 in the order P1 (burst 8), P2 (burst 4), P3 (burst 2). The scheduler is non-preemptive shortest-job-first. ~~The machine uses a fast SSD.~~ What is the average waiting time?

Target word: average waiting time under SJF **Decides:** SJF runs P3, then P2, then P1.

A	Adjacent category	FCFS order: $(0 + 8 + 12) / 3$.
B	Confusion set	Average turnaround time: $(2 + 6 + 14) / 3$.
C	Correct	Waits: P3 0, P2 2, P1 6; $(0 + 2 + 6) / 3 = 2.67$.
D	Dropped element	Total waiting time, not divided by 3.

4. Answer D Concept 2 · transfer

Under first-come, first-served scheduling, jobs P1 (burst 8), P2 (burst 4), and P3 (burst 2) arrive at time 0 in that order. What is P3's waiting time?

Target word: single job's wait under FCFS **Decides:** P3 starts after P1 and P2 finish.

A	Confusion set	P3's turnaround time.
B	Wrong stem number	P3's burst time.
C	Inverted operation	SJF would run P3 first; FCFS runs it last.
D	Correct	$8 + 4 = 12$.

5. Answer D Concept 3

An administrator sets a round-robin time quantum of 1 ms on a system where each context switch takes 0.5 ms. ~~The machine hosts shells for 200 users.~~ What is the main effect?

Target word: effect of a tiny quantum **Decides:** Switch cost is half the quantum.

A	Inverted operation	That happens when the quantum is very large.
B	Camouflage	User count does not decide switch overhead.
C	Absolute language	Overhead grows as the quantum shrinks.
D	Correct	$0.5 / (1 + 0.5) \approx 33\%$ overhead.

6. Answer B Concept 4

Job A holds a lock on the Orders table and waits for Customers; Job B holds Customers and waits for Orders. ~~Both run nightly at 2 a.m. on a server with 64 GB of RAM.~~ The fix requires every job to take locks in the same fixed order. Which deadlock condition does this break?

Target word: condition broken **Decides:** A global lock order prevents a cycle.

A	Confusion set	Locks are still exclusive.
B	Correct	With a fixed order, no cycle of waiting can form.
C	Camouflage	Memory does not break a lock cycle.
D	Adjacent category	Jobs still hold one lock while waiting for the next.

7. Answer C Concept 4 · transfer

A system requires every process to request all the resources it will need at once, before it starts. Which deadlock condition does this remove?

Target word: condition — reversed from rule **Decides:** No process holds some resources while waiting for others.

A	Absolute language	Only one condition is removed.
B	Adjacent category	Broken by lock ordering, not all-at-once requests.
C	Correct	A process either gets everything up front or holds nothing.
D	Confusion set	Broken by allowing resources to be taken away.

8. Answer A Concept 5

A system uses 4 KB (4,096-byte) pages ~~and has 16 GB of RAM.~~ What are the page number and offset for logical address 13,000?

Target word: page number and offset **Decides:** $13,000 \div 4,096 = 3$ remainder 712.

A	Correct	$3 \times 4,096 = 12,288$; $13,000 - 12,288 = 712$.
B	Inverted operation	Swaps quotient and remainder.
C	Dropped element	Drops the subtraction for the offset.
D	Confusion set	Rounds up and measures back from the next page boundary.

9. Answer A Concept 5 · transfer

With 1,024-byte pages, page 5 maps to frame 9. What physical address matches offset 100 in page 5?

Target word: solve for physical address **Decides:** Physical = frame × page size + offset.

A	Correct	$9 \times 1,024 + 100 = 9,316$.
B	Inverted operation	Uses the page number instead of the frame.
C	Wrong stem number	Uses 1,000 instead of 1,024 bytes per page.
D	Dropped element	The frame's start, without the offset.

10. Answer A Concept 6

After an administrator raised concurrent batch jobs from 10 to 40, CPU use fell from 70% to 15% and the disk is busy almost entirely with paging. ~~The server came from a reputable vendor last year.~~ What is happening, and what is the fix?

Target word: diagnose and fix **Decides:** More jobs, low CPU, heavy paging.

A	Correct	Too few frames per process causes constant paging.
B	Camouflage	The vendor detail does not decide the cause.
C	Adjacent category	The CPU is waiting on paging, not overloaded.
D	Inverted operation	More jobs deepen the thrashing.

11. Answer D Concept 7

Two threads each run $\text{balance} \leftarrow \text{balance} + 100$ on a shared account 1,000 times with no locking. The expected total increase is 200,000, but the actual increase is lower. ~~The app runs on a four-core laptop.~~ What is the fix?

Target word: fix lost updates **Decides:** Unprotected read-modify-write on shared data.

A	Absolute language	Threads are safe with proper synchronization.
B	Dropped element	Makes collisions rarer but does not prevent them.
C	Camouflage	More cores make races more likely, not less.
D	Correct	A critical section stops interleaved reads and writes.

12. Answer B Concept 8

A print-spooler service runs with full administrator rights on every workstation, though it only needs to write to the print-queue folder. ~~The service vendor has been in business for 20 years.~~ What is the best control?

Target word: defensive control **Decides:** The service has far more rights than it needs.

A	Adjacent category	Authentication strength does not limit what the service can do.
B	Correct	Least privilege limits the damage any flaw can cause.
C	Camouflage	Vendor age does not reduce the exposure.
D	Absolute language	Breaks needed functions.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Processes vs. threads	Threads of one process share its address space (heap, globals, open files); separate processes have isolated memory.	Thinking threads have private copies of memory, or that processes share it.	A shared cache, a crash, or isolation between tabs or workers.
2	CPU scheduling and waiting time	Waiting time = start time – arrival time (non-preemptive); shortest-job-first minimizes average waiting time.	Confusing waiting time with turnaround time, or FCFS with SJF order.	Burst times for jobs arriving together.
3	Round-robin time quantum	A quantum too small wastes CPU on context switches; too large makes round robin behave like FCFS.	Believing smaller is always better.	A quantum close to the context-switch cost.
4	Deadlock conditions	Deadlock needs mutual exclusion, hold and wait, no preemption, and circular wait; prevention breaks one.	Matching the wrong condition to a prevention rule.	A lock-ordering or all-at-once allocation rule.
5	Paging address translation	Page number = address ÷ page size (integer); offset = remainder; physical = frame × page size + offset.	Using the page number in place of the frame, or dropping the offset.	A page size and a logical address.
6	Thrashing	When processes lack enough frames, the system spends its time paging and CPU use collapses; reduce multiprogramming or add memory.	Adding more jobs because the CPU looks idle.	Low CPU plus heavy page-in/page-out after raising the job count.
7	Race conditions and mutual exclusion	Shared read-modify-write sequences must run inside a critical section protected by a lock.	Adding delays or cores instead of synchronization.	Two threads updating one shared value and a total that comes up short.
8	Least privilege	Run each service with only the rights it needs so a flaw in it cannot reach the whole system.	Relying on vendor trust or passwords in place of limiting privilege.	A service running as administrator that needs one folder.

Course scope (public description): Operating Systems for Computer Scientists focuses on the intricacies of operating systems. This comprehensive course for computer science students covers core principles such as processes, threads, memory management, and file systems, providing students with insights into CPU scheduling algorithms, deadlock handling, and system performance optimization. Additionally, the course delves into security mechanisms, addressing common threats and preventative measures. Through a blend of theoretical concepts and practical applications, students emerge equipped to adeptly navigate operating system features and prepared for real-world challenges in computer science.